Sangbum Daniel Choi

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Summary

Former entrepreneur and research scientist with 6+ years of experience leading projects across multimodal AI, on-device inference, AI agents, and production ML systems. Co-founded a startup and served as CTO, led teams of up to eight people, and delivered projects from research through deployment. Extensive open-source contributor across the Hugging Face ecosystem and creator of OpenLocalAgent, OCR, and Absorbed.

Experience

- **Toss Bank** — Seoul, South Korea
  Data Scientist — *Jan 2026 - Present*
  - [**Lead**] **AI Authentication**: Developed a mobile ID-card recognition and facial verification model for real-time on-device inference. Validated deployment across numerous mobile devices; combined face anti-spoofing, multitask depth/normal/keypoint estimation, and face-embedding similarity using ONNX models and WebGPU.
  - **Language Model Evaluation**: Built an in-house benchmark aligned with public evaluation protocols and scoring heuristics to compare Qwen3.5-397B, Qwen3.8B, GLM-5.3, and DeepSeek V4 Flash 0731, focusing on agentic capability and Korean-language understanding.
  - **On-premise AI Answering Agent**: Built a Qwen3-0.6B tool-calling agent as an internal Slack bot, routing requests to direct answers or internal tools. Integrated it with the LLM gateway and automated session refresh.
  - **On-premise AI Coding Agent**: Built coding-agent workflows with GLM-5.3 and Qwen3.5-397B across Claude Code-assisted experimentation, OpenCode CLI/TUI, Mac PVDI, notebooks, and Kubeflow. Designed planner/critic/executor orchestration with lean and heavy execution tiers, routing prompts to shell or Kubeflow tools for code and file operations and job submission, status, and cancellation.
  - [**Lead**] **End-to-End Document AI**: Led the full research and production pipeline for document extraction, combining an approximately 1B-parameter vision-language model for multi-page classification, orientation, layout detection, and OCR/table understanding with a 35B-parameter MoE LLM (3B active parameters) for key-value extraction; post-trained the VLM, designed stage-level diagnostics, and reached 61% exact-match accuracy for automation-ready outputs.
- **SuperbAI** — Seoul, South Korea
  Machine Learning Engineer — *Sep 2021 - Jan 2026*
  - [**Lead**] **Multimodal Foundation Model**: Held approximately one year of team-leadership responsibility, leading two ML engineers through multimodal pre-training and staged text/image alignment for a visual-grounding model; curated 1.1M images, then drove evaluation and deployment.
  - **CVPR 2025 Competitions**: Won 2nd place in IOD and 4th place in FSOD challenges, validating the model on open evaluation benchmarks.
  - [**Lead**] **ML Training & Serving Platform**: Built reusable distributed multi-GPU training, evaluation, and deployment infrastructure with PyTorch and AWS Batch; optimized production serving with TensorRT/Triton, improving throughput by 5x and delivering 1,100+ models and 60+ customer endpoints in one year.
  - **Post-training and Efficiency**: Applied LoRA, Adapter, and LST methods and memory-efficient training, reducing GPU memory by 65.6% and training time by 44.3% while preserving model performance.
  - **Interactive Segmentation**: Built browser, hybrid, and cloud-based segmentation tools using RepViT-SAM, FocalClick, SAM, and SAM2, increasing segmentation labeling speed by 25x.
- **Kakao Enterprise** — Pangyo, South Korea
  Machine Learning Engineer Intern — *Feb 2021 - Aug 2021*
  - **AutoML**: Fixed AutoGluon NeuralNetFastAI scaling issue and developed a Flask-based training/inference AutoML framework with a simple front-end.
- **Team ISLAND** — Seoul, South Korea
  Chief Technology Officer — *Mar 2019 - Oct 2020*
  - **Startup**: Co-founded the company as CTO, raised approximately $290K, and managed a five-developer core team across Android, Unity, and deep learning; across leadership roles, led up to eight people simultaneously.
  - **Mobile ML**: Built ZZAZZ, a mobile video editor using detection/segmentation, 3D mapping, and tracking; deployed lightweight vision models through TensorFlow Lite.

Open Source

- **OpenLocalAgent**: Building small tool-calling agents from scratch in pure PyTorch using open data, with pretraining, midtraining, post-training, and evaluation across ten public agent benchmarks.
- **OCR**: Building a reproducible evaluation and improvement framework for sub-1B document VLMs, covering accuracy, calibration, robustness, and LoRA adaptation.
- **Absorbed**: Developing an open-source SwiftUI roguelike with turn-based combat, branching expeditions, spirit capture, persistent progression, and English/Korean localization.
- **Hugging Face Transformers**: Contributed 40+ pull requests to a large production codebase across multimodal and vision models, independently shipping model architectures, processors, conversion scripts, distributed training fixes, tests, and documentation.
- **Hugging Face Transformers - SAM2**: Led the addition of Segment Anything 2 support to Transformers, including model integration, image/video processors, conversion flow, documentation, and tests; credited in the official SAM2 docs.
- **Hugging Face Transformers - Molmo2**: Opened Molmo2 support for Transformers and published danelcsb/Molmo2-4B on the Hugging Face Hub.

Papers

- **ZERO: Multi-modal Prompt-based Visual Grounding**: arXiv - Jul 6, 2025
- **MobileHumanPose: Toward real-time 3D human pose estimation in mobile devices**: IEEE/CVF CVPR Workshop - Jun 19, 2021
- **Improved DOA estimation of MUSIC algorithm based on irregular microphone array**: UIUC Research - Dec 20, 2018
- **On Joint Transfer of Energy and Information: A Markov Decision Problem Formulation**: Research paper - Dec 13, 2018
- **Terahertz Signal Analysis in Tissue of Gastroesophageal Reflux Disease**: Sejong Science High School - Dec 31, 2013

Education

- **KAIST (Korea Advanced Institute of Science and Technology)** — Daejeon, South Korea
  Professional Master in Electrical Engineering; GPA: 3.55/4.3 — *Feb. 2020 – Aug. 2021*
- **POSTECH (Pohang University of Science and Technology)** — Pohang, South Korea
  Bachelor in Electrical Engineering; GPA: 3.4/4.3 — *Mar. 2015 – Feb. 2020*
- **UIUC (University of Illinois Urbana-Champaign)** — Illinois, United States
  Exchange Student; GPA: 3.6/4.0 — *Jan. 2018 – Dec. 2018*

Programming Skills

- **Research**: Python, PyTorch, Transformers, multimodal pre/post-training, LoRA, contrastive alignment, evaluation
- **Systems**: Linux, AWS Batch, multi-GPU training, MLflow, TensorRT, Triton, ONNX, TensorFlow Lite, Git
- **Agents**: LangChain, Langfuse, Claude Code, Codex, opencode
- **Languages**: Korean (native), English (professional)